where $\boldsymbol{\varphi}$ is an l-dimensional vector whose kth component is

$$\varphi_k = \boldsymbol{p}(t)^T \boldsymbol{M}^{-1} \nabla^2 g_k(\boldsymbol{q}(t)) \boldsymbol{M}^{-1} \boldsymbol{p}(t).$$

Using (4.8) in (4.10) we obtain

$$\boldsymbol{g}'(\boldsymbol{q}) \boldsymbol{M}^{-1} [\boldsymbol{F} - \boldsymbol{g}'(\boldsymbol{q})^T \boldsymbol{\lambda}] + \boldsymbol{\varphi} \equiv \boldsymbol{0},$$

This is a system of linear equations for $\boldsymbol{\lambda}$. Assuming $\boldsymbol{g}'(\boldsymbol{q}) \boldsymbol{M}^{-1} \boldsymbol{g}'(\boldsymbol{q})^T$ is invertible, we may solve for $\boldsymbol{\lambda}$. Reinserting this expression for $\boldsymbol{\lambda}$ into (4.8) then gives a system of ordinary differential equations whose solutions, starting from a point on $T^*\mathcal{M}$, will remain on $T^*\mathcal{M}$ for all time. The co-tangent bundle becomes an invariant manifold of this ODE system. $\boldsymbol{\lambda}$ is defined at any time by the physical variables $\boldsymbol{q}$ and $\boldsymbol{p}$.

The presence of such an underlying ordinary differential equation is very useful for conceptual purposes, but in practice it is usually found to be preferable to work directly with the constrained formulation (4.7)–(4.9) for numerical discretization and for describing statistical mechanical properties.

The formulation of Hamiltonian systems with constraints is discussed from various perspectives in [15, 151, 249]. Here, we take a practical perspective on these systems, addressing ourselves only to the issues that are directly relevant for the extension of numerical methods derived in previous chapters to the constrained setting. More detailed treatment of numerical methods for constrained mechanical systems may be found in [164, 227].

4.3.1 Symplectic Structure with Holonomic Constraints

Describing the Hamiltonian structure for a constrained system is a little complicated to do formally. The simplifying concept that we exploit is that the symplectic 2-form in the ambient space can be projected to the co-tangent bundle to define an associated symplectic form on the manifold.

One way to understand the symplecticness of the Hamiltonian flow in the unconstrained case is as follows. Think of the flow map as defining $\boldsymbol{Z}(t, z) = \mathscr{F}_t(z)$, where we can write the vector $\boldsymbol{Z}(t, z) = [Z_1(t, z), \ldots, Z_{2N_c}(t, z)]$. Then the symplecticness condition is

$$\mathrm{d}\boldsymbol{Z} \wedge \boldsymbol{J} \mathrm{d}\boldsymbol{Z} = \mathrm{d}z \wedge \boldsymbol{J} \mathrm{d}z.$$

Taking the time derivative, as the right hand side is constant we have

$$\mathrm{d}\boldsymbol{Z} \wedge \boldsymbol{J} \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}\boldsymbol{Z}) + \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}\boldsymbol{Z}) \wedge \boldsymbol{J} \mathrm{d}\boldsymbol{Z} = 0,$$